

Master's Programme in ICT Innovation

Enhancing Thematic Analysis with Large Language Models: A Comparative Study of Structured Prompting Techniques

Leveraging Large Language Models to Automate and Enhance Inductive Thematic Analysis in Qualitative Research

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Abstract

Thematic analysis, while crucial for qualitative research, becomes challenging with large datasets due to the intensive manual effort involved. This thesis explores the capacity of Large Language Models (LLMs) to automate and enhance this process through different prompting techniques, benchmarking their performance against human-generated analyses. In this thesis we compare three distinct approaches: (1) Zero-Shot prompting, a single-prompt solution relying solely on pre-trained knowledge; (2) Top-Down Structured Prompting (TDSP), leveraging prompt engineering principles by starting with broad themes and progressively refining analysis; and (3) Bottom-Up Structured Prompting (BUSP), mirroring the established six steps of human thematic analysis designed for efficient extraction of nuanced insights. Applying these techniques to three diverse qualitative datasets, we used a rubric-based scoring system to evaluate the outputs against findings from the original research. Our results demonstrate that structured prompting techniques, particularly TDSP, significantly outperform the Zero-Shot approach in accurately capturing key findings. On average, 86% of findings identified by human researchers were at least partially identified in the outputs generated using structured prompting techniques. We also explored using an AI-based evaluator for rubric scoring but found the results unreliable, indicating that human evaluation remains crucial. This study makes a compelling argument that AI can efficiently perform thematic analysis and, especially in the future, provide substantial support to qualitative researchers in their work.

Keywords Large Language Models, Thematic Analysis, Qualitative Research, Artificial Intelligence, Prompt Engineering, Inductive Thematic Analysis

1 INTRODUCTION

The advent of large language models (LLMs), especially since the release of ChatGPT 3.5, has sparked a surge of interest in leveraging their capabilities across various domains. From generating creative text formats [10] to solving complex reasoning problems [43], LLMs have demonstrated remarkable versatility and have achieved impressive performance on benchmark tasks like SQuAD 2.0 for reading comprehension [28], DROP for reasoning over paragraphs [2], and DocBench for document understanding [48]. However, LLMs, much like human analysts, face challenges when confronted with large volumes of text. Their performance often degrades as the amount of information they need to process increases [27], potentially hindering their ability to effectively handle tasks that require a deep understanding of complex, nuanced data.

Qualitative research, a cornerstone of social science and humanities, relies on analyzing precisely this type of data that is rich, has detailed narratives and is gathered through methods like interviews, focus groups, and open-ended surveys [20]. The goal is to explore complex human experiences, behaviors, and social phenomena in depth [4, 20, 25]. Thematic analysis is a widely recognized method in qualitative research for systematically identifying present patterns, known as themes, inside the qualitative data [6, 7]. These themes represent core concepts or ideas that emerge from the data. Researchers first identify key insights (codes) within the data that relate to the research question. These codes are then grouped and organized into broader themes, which are finally synthesized into a comprehensive evaluation that connects the findings to the overall research goals [7]. While effective, thematic analysis is often time-consuming and cognitively demanding, especially with large datasets, posing a significant challenge for researchers.

This study demonstrates the potential of combining human-inspired methodologies, like thematic analysis, with advanced prompting techniques from the field of prompt engineering to enhance the performance of LLMs in automating this demanding task. By adapting the structured framework of thematic analysis and applying techniques like successive prompting and context optimization [16, 27], we aim to guide LLMs towards a more methodical and nuanced understanding of qualitative data. We introduce two distinct models built on these principles, both classified as Structured Prompting Techniques (SPT): Top-Down Structured Prompting (TDSP) and Bottom-Up Structured Prompting (BUSP). TDSP leverages the LLM’s strengths in pattern recognition by starting with broad themes and progressively refining the analysis to extract key insights. In contrast, BUSP mirrors the process of human-driven thematic analysis, beginning with detailed code generation and building up to the identification of overarching themes. We will then manually evaluate their performance. This research is particularly timely and relevant, as it moves beyond evaluating LLMs on smaller, benchmark-focused tasks and explores their capacity to handle the complexities of a real-world, human-centered research process.

2 RESEARCH PROBLEM

One of the first approaches one might envision for using Large Language Models in Qualitative Research is to simply paste the entire interview dataset into the model and ask it to generate a summary of key findings. This approach, called zero-shot prompting, has been shown to produce worse results than more advanced techniques like chain-of-thought prompting, which guides the model through the task at hand [43]. This revelation symbolizes the idea behind prompt engineering and opens the discussion on how guiding an LLM can lead to better results.

While thematic analysis holds potential as a guide for LLMs, it remains unexplored whether these models truly benefit from this approach and if further prompt engineering can yield even better results.

2.1 Research Objective

This research investigates the potential of Large Language Models (LLMs) to perform autonomous thematic analysis on qualitative data. Our primary objective is to evaluate whether guiding LLMs through a structured, multi-step process, as opposed to a single-prompt approach, enhances the depth, nuance, and accuracy of their generated insights. Specifically, this study aims to:

- (1) Assess the effectiveness of a single-prompt for generating a thematic evaluation, establishing a baseline for LLM capabilities.
- (2) Evaluate the effectiveness of structured prompting that decomposes thematic analysis into distinct steps, mirroring the process used by human researchers.
- (3) Conduct a comparative analysis of the results generated by both LLM approaches and those produced by human researchers in the original studies, by examining the insights derived from the different evaluations.
- (4) Test if LLMs can be used to evaluate their own thematic analysis results, potentially automating the evaluation process.

By pursuing these goals, this study seeks to enhance our comprehension of how AI can be effectively incorporated into qualitative research methods, potentially leading to more reliable and insightful automated thematic analysis tools and overcoming the current limitations of LLMs in handling complex tasks.

2.2 Research Question

The central research question guiding this study is:

To what extent do structured, multi-step prompting techniques, mimicking the stages of human thematic analysis, enhance the depth and nuance of insights generated by Large Language Models in qualitative data analysis, compared to simpler prompting approaches and human-generated analyses?

3 ETHICAL CONSIDERATIONS

3.1 Bias in LLMs

A key ethical concern when using LLMs for thematic analysis is their potential for bias. Trained on massive datasets, LLMs can reflect and magnify existing biases present in the source material [5]. This can result in skewed thematic analyses that over-represent dominant perspectives and marginalize others. As [Bender et al. \(2021\)](#) caution, trusting the output of LLMs can give a false sense of correctness, even when the underlying data the models were trained on is already biased. Therefore, researchers using LLMs for qualitative research must be vigilant in mitigating this inherent bias through careful data selection, meticulous prompt engineering, and critical evaluation of AI-generated results.

3.2 Impact on Employment and Human Oversight

While AI can automate aspects of thematic analysis, it does not eliminate the need for human researchers. Instead, AI tools should be seen as powerful assistants, freeing researchers to focus on higher-level analysis and interpretation. This shift requires researchers to adapt to new roles emphasizing oversight and

critical evaluation of AI-generated output. Human intervention remains crucial to ensure the accuracy, relevance, and freedom from bias in the final thematic analysis.

3.3 Sustainability of LLMs

In considering the ethical implications of Large Language Models (LLMs) for qualitative research, the issue of sustainability must be addressed. LLMs, particularly those at scale, require significant computational power, which translates into high energy consumption. The Morgan Stanley Cooperation predicts that by the year of 2027, the LLM energy consumption will increase each year by 70% [3]. Furthermore, the usage of larger models has shown to significantly increase power consumption per generated word [41]. While this study utilizes ChatGPT-4o and Gemini Pro, two of the largest commercially available models, we acknowledge the importance of evaluating whether any of the prompts used in our study could be handled by smaller models. This would enhance the sustainability of LLM-driven thematic analysis, particularly for future commercial applications. Finally, we believe that using LLMs can enhance the quality of research, ultimately benefiting humanity as a whole. However, users should remain mindful of the environmental impact associated with the use of such methods.

4 LITERATURE REVIEW

4.1 Thematic Analysis in Qualitative Research

Thematic analysis is a foundational method in qualitative research for "identifying, analyzing, and reporting patterns (themes)" within data [7]. It offers a flexible approach to analyzing qualitative data, allowing researchers to distill large datasets into meaningful insights. This method is widely used across various research fields due to its ability to systematically explore and interpret complex data, making it an essential tool for qualitative researchers.

Generally thematic analysis can be broken down into the following steps. In the first step the researchers agree on a set of codes that summarize interesting ideas or concepts in the data, which is called a codebook. In the next step the codebook is applied to the data, allowing the data to be broken down into subsets only responding to a specific code. The researchers can then group the codes into themes and subthemes, guiding a structure for the final evaluation.

4.2 Thematic Analysis Approaches

Thematic analysis can generally be classified into two main approaches: grounded theory and deductive qualitative research [8]. While deductive analysis works with pre-existing theories or frameworks, grounded theory is based on the idea that all results should emerge from the underlying data. Inductive thematic analysis is a specific approach of grounded theory where themes emerge directly from the data, by following a structured analysis approach [6]. Inductive thematic analysis is particularly useful when exploring new research areas or when research questions are open-ended. All of the datasets used in our study employ inductive thematic analysis, while one of the dataset uses a more specialized version of inductive thematic analysis called reflexive thematic analysis [15, 32, 38]. Reflexive thematic analysis is similar to inductive thematic analysis in that it is also grounded in the data, but it acknowledges the researcher's biases and reflections [8]. Braun and Clarke argue that by acknowledging the researcher's knowledge about the topic and their own biases, thematic evaluation will yield better results. Reflexive thematic analysis has since been widely adopted in the social sciences, [34]. In this paper, we cannot

simulate researcher bias, but since our datasets are grounded in reflexive and inductive thematic analysis, we will use the different definitions for inductive and reflexive thematic themes and codes. Reflexive themes focus on a key idea or learning from the dataset, while inductive themes are similar but allow for a theme to summarize a topic [8].

4.3 6 Steps of Thematic Analysis by Braun & Clarke

For our SPT we will base our research on the six steps of Thematic Analysis as proposed by Braun and Clarke, which are widely recognized and accepted as a template for conducting thematic analysis [7].

- (1) **Familiarization with the Data:** Researchers begin by immersing themselves in the data, reading and re-reading it to note initial observations. This step is essential for understanding the scope and content of the dataset.
- (2) **Generating Initial Codes:** The data is systematically coded, with segments tagged with labels that capture important features relevant to the research questions. This process breaks down the data into manageable pieces, facilitating the identification of recurring patterns.
- (3) **Searching for Themes:** Codes are then analyzed to identify broader patterns of meaning, or themes. These themes represent significant aspects of the data aligned with the research objectives. This step involves organizing codes into potential themes, much like clustering related data points in an algorithm.
- (4) **Reviewing Themes:** The identified themes are refined and evaluated for coherence, ensuring they accurately reflect the data and are consistent with both the coded extracts and the entire dataset. This quality control step ensures the robustness and relevance of the themes.
- (5) **Defining and Naming Themes:** Each theme is clearly defined, with a narrative that explains its significance and how it relates to the research questions. This step ensures that the themes are distinct and contribute to a comprehensive understanding of the data.
- (6) **Producing the Report:** Finally, the themes are woven together into a coherent narrative that answers the research questions. This report illustrates how the themes are supported by data extracts and how they contribute to the overall analysis, much like compiling the output of an algorithm into a final, interpretable format.

During our research we will test if forcing an LLM to follow a variation of those 6 steps of Thematic Analysis can improve the results of qualitative research done on interview Datasets by LLM's.

4.4 Large Language Models

Large Language Models (LLMs) like Google Gemini 1.5 and ChatGPT-4 signify a substantial advancement in artificial intelligence. These models are trained on exhaustive datasets consisting of multiple different data inputs, enabling them to generate, understand, and manipulate text with remarkable sophistication, mirroring human communication [28]. LLMs excel in various language tasks, ranging from basic text generation to complex problem-solving [2, 28, 48].

4.5 Prompt Engineering

While LLM's have shown exceptional capabilities prompt engineering has emerged as a tool to further increase the performance of the models. Therefore prompt engineering has emerged as an important tool to enhance the performance of generalist models like ChatGPT, since it leads to better results, whilst keeping the generalist nature of the LLM [40]. This process entails creating tailored instructions, known

as prompts, to direct the LLM’s responses [36]. Similar to how humans respond differently to rephrased questions, LLMs generate varied outputs based on prompt structure and context. We will now look at different prompting techniques that have been developed to enhance the performance of LLM’s.

Zero-shot prompting, the simplest form, provides the LLM with a task description without examples [40]. While efficient, it struggles with complex tasks [12]. Few-shot prompting enhances performance by providing input-output examples, demonstrating the desired behavior [10, 45]. However, its effectiveness diminishes with increasing task complexity, particularly for tasks demanding intricate reasoning steps [43].

Chain-of-Thought (CoT) prompting [43] encourages more sophisticated reasoning by prompting the LLM to come up with a chain of steps that result in the final answer. This method excels in arithmetic tasks, where questions can be segmented into sequential calculations. CoT prompting also significantly improves reasoning tasks on benchmarks like StrategyQA, which tests reasoning abilities with questions like “Did Aristotle use a laptop?” [18, 43]. CoT correctly answers approximately 10% more questions than few-shot prompting.

Inspired by the success of CoT prompting in guiding LLMs through reasoning steps, researchers explored whether further subdividing these reasoning steps into smaller, more manageable subtasks could unlock even greater LLM capabilities, similar to how humans break down complex problems. This led to the development of “structured prompting” techniques. Notable examples include Least-To-Most prompting, which creates increasingly complex subtasks, successive prompting, which generates a plan of subtasks, and decomposed prompting [16, 23, 47]. Decomposed Prompting draws inspiration from proven problem-solving strategies commonly used in fields like software engineering. It involves decomposing a task into sub-tasks aligned with strategies such as recursion, looping, and divide-and-conquer. By framing the problem in a familiar and structured manner, Decomposed Prompting aims to leverage the LLM’s ability to learn and apply these established problem-solving approaches [23]. Decomposed prompting has shown to lead to better results than CoT Prompting on all complex reasoning tasks it was tested on, giving us a strong belief that it can be beneficial for thematic analysis as well [23].

Building upon the understanding that structured prompting can improve LLM performance on complex reasoning tasks, particularly by reusing proven problem-solving skills, this thesis introduces the Structured Thematic Prompting (STP). Inspired by Braun and Clarke’s six-step framework for thematic analysis [7], this structure guides LLMs through a systematic process mirroring thematic analysis in qualitative research. By decomposing this complex task into smaller, manageable sub-tasks, the structure allows the LLM to focus its abilities on specific aspects of the process, potentially leading to more accurate, coherent, and insightful thematic analyses of textual data.

4.6 Context Window

Large language models operate within the constraints of a “context window,” which limits the amount of information they can process at once. This window encompasses both input (the prompt and any provided context) and output (the model’s generated text). For instance, ChatGPT-4o has a substantial input context window of 128,000 tokens (one token is roughly equivalent to one word), while its output is limited to 4,096 tokens [1]. Google’s Gemini Pro has an even larger input capacity of 2 million tokens but maintains a similarly small output limit of 8,192 tokens [35].

These limitations, especially on output length, necessitate a strategic approach for complex analysis tasks like thematic evaluation. Such tasks often demand the breakdown of the overarching analysis into smaller, more manageable sub-tasks that fit within the context window.

Furthermore, research indicates that longer prompts can lead to less accurate results [12, 27]. Excessive input can cause the model to lose focus, potentially diverging from the intended question or overlooking crucial information. Conversely, information presented at the beginning and end of the input tends to have a higher influence on the output [26]. These observations suggest that minimizing the context window, whenever possible, can enhance the accuracy and focus of LLM outputs. This aligns with the principles of thematic analysis outlined by Braun and Clarke, where initial coding helps researchers focus on specific aspects of their data. By sorting the data by codes, researchers can then engage more deeply with each theme, effectively reducing their own "context window" [7]. Our proposed Inductive Thematic Prompting Structure leverages this concept to guide LLMs towards more focused and insightful thematic analysis.

4.7 LLM Agents

LLM agents represent an advanced application of large language models, moving beyond traditional use cases like zero-shot or few-shot learning. Unlike basic scenarios where the model simply responds to isolated inputs, LLM agents are designed for complex, dynamic interactions. They incorporate advanced control structures such as planning, looping, context optimization, and reflection, enabling them to perform comprehensive tasks ranging from data analysis to generating long-form content [12][29].

These agents leverage LLMs' inherent reasoning capabilities, allowing them to adapt their strategies based on ongoing analysis and understanding of the data. The inclusion of functionalities like tool use and function calling further extends their utility, enabling LLM agents to engage in tasks that require sustained and evolving interactions [29, 29]. Another important aspect of LLM agents is context optimization, letting the agent autonomously select context for successive prompts, optimizing performance of the model [26]. This adaptability makes LLM agents powerful tools for conducting detailed qualitative research, where the ability to iterate and refine analysis is crucial.

While the STP technique presented in this thesis does not constitute a fully autonomous LLM agent, it exhibits certain agent-like characteristics. The decomposition of the thematic analysis task into sub-tasks, with the output of one feeding into the next, echoes an agent's planning and execution of a multi-step process. Then each step in the process has a predefined reduced context, optimizing performance. Also the AI is utilized to self select the most relevant results and lastly, we perform post processing on results, which can be compared to tool use of LLM agents. However, the current structure lacks certain crucial elements of true LLM agents, notably autonomous planning of which step to perform and the evaluation of its own results [29]. Nevertheless, by incorporating these agent-like features, the Inductive Thematic Prompting Structure demonstrates the potential of borrowing from the agent paradigm to enhance autonomous thematic analysis.

4.8 LLMs in Thematic Analysis

The integration of Large Language Models (LLMs) into qualitative research and especially thematic analysis (TA) is a rapidly developing area within qualitative data analysis. Initial research demonstrates the potential of these advanced language processing systems to contribute to various stages of the TA process [14, 46].

When provided with pre-defined topics to look for in the data (the codebook) as in deductive Thematic Analysis, LLMs have shown promising capabilities in replicating human coding practices. One study has shown this by comparing the thematic codes being applied by a LLM and a researcher to a interview dataset, where they have reported moderate levels of intercoder agreement between LLMs and human coders, with values reaching around 0.65 [46]. The intercoder agreement present the overlap between two researchers application of codes to the data. A higher value here indicates that the two researchers applied more of the same codes to distinct data segments [24], with a value of 0.65 indicating a substantial agreement. This indicates that LLMs are capable of accurately labeling pre-established codes for qualitative data, potentially increasing the efficiency and consistency of the coding process.

Beyond applying the codebook, researchers are exploring the potential of LLMs to contribute to codebook generation and theme mapping in inductive thematic analysis (ITA). In ITA the codebook is the set of codes that summarize the key findings inside the dataset, while the themes structure the codes into sets of shared meaning. Current research is investigating various human-LLM collaboration models to enhance codebook development. Chew et al. proposed the concept of "Large Language Model-assisted Content Analysis" (LACA), where LLMs actively contribute to both codebook co-development and coding tasks [13]. Tools like the "CoAICoder" [17] facilitate human-LLM collaboration by offering code suggestions and identifying potential coding discrepancies. Taking a step towards more autonomous TA, De Paoli demonstrated the feasibility of using an LLM to generate themes similar to those produced by human researchers [14]. Their approach involved a simplified thematic analysis process, prompting the LLM to generate potential themes for each interview individually, which were then aggregated and organized into a thematic framework. In our research we want to go a step further and not just investigate if LLM's can generate themes, but if they can generate concise findings similar to those produced by human researchers.

4.9 Rubric Scoring

Rubric scoring [9] is a valuable tool for us to evaluate the output of generated results by our models in comparison to results from human researchers on the same dataset. While rubric scoring has found applications across various fields, its origins lie in the field of education, where it emerged as a method for evaluating student work in a more structured and objective way. As Brookhart states, scoring rubrics, regardless of their specific form, offer significant benefits in terms of reducing bias and ensuring a more consistent and transparent evaluation process [9].

Essentially, a rubric functions as a detailed scoring guide that outlines the specific characteristics associated with different levels of performance for a given task or subject [9, 31]. Instead of relying solely on subjective impressions, evaluators can refer to the rubric to make judgments based on pre-defined criteria and performance levels. For example, a rubric might define distinct criteria for classifying a statement as "fully fulfilled," "partially fulfilled," or "not fulfilled". By clearly defining those categories evaluators are more likely to agree in their assessment. This reduces the potential for personal biases to influence the evaluation and ensures that all subjects are assessed according to the same standards. These rubrics can either be tailored to a specific task, known as a task-specific rubric, or designed to assess a broader skill set or category of tasks, making them applicable across multiple assessments [31]. This latter type is referred to as a general rubric. Regardless of whether a rubric is task-specific or general, the core benefits of using a rubric remain consistent: increased objectivity, transparency, and consistency in evaluation.

Rubric Scoring is of interest in our study, as we can utilize a rubric scoring table for setting quantifiable metrics to analyze the output of the different STPs. By defining rubrics for each finding we expect to find, we can objectively determine if it was found in the proposed output of the LLM.

4.10 LLMs in Reading Comprehension and Rubric Scoring

As specified in the Research Objective another important aspect of this research is to evaluate if LLMs can be used to evaluate their own thematic analysis results. This topic was important to us as the research suggest promising results for self-evaluation.

Recent advancements in Large Language Models (LLMs) have resulted in remarkable progress in reading comprehension. These models are now capable of effectively processing and understanding complex documents, accurately answering related questions. Benchmarks such as DocBench, created to assess the performance of LLM-based document reading systems, demonstrate this remarkable capability. DocBench findings highlight almost perfect overlap of around 97% between human annotators and LLMs in answering questions based on provided documents, indicating a strong understanding of the textual content [48]. Further solidifying these advancements, GPT-4, a leading LLM, achieved an accuracy of 83% on the DROP benchmark, which focuses on complex reasoning over paragraphs, requiring models to go beyond simple fact retrieval [2].

This proficiency in reading comprehension makes LLMs promising candidates for rubric-based evaluation, and researchers have suggested using LLMs for rubric based student scoring [45]. The ability to accurately interpret information within a document directly translates to the capability of judging a subject against pre-defined criteria within a rubric. The paper "Evaluating the Evaluator: Measuring LLMs' Adherence to Task Evaluation Instructions" provides insightful evidence of this potential [33]. The study found that providing LLMs with detailed rubric information, similar to what's found in a well-defined rubric, generally leads to slightly better alignment with human judgments, even though the gains are relatively small. This suggests that rubric scoring can guide LLMs in making slightly more accurate evaluations. Specifically, GPT-4 demonstrated high correlation with human judgments when evaluating criteria like completeness, achieving a correlation of 0.764, but only a moderate correlation on relevance, meaning how well the content of a solution directly addresses the specific requirements of the given task [33]. This indicates that LLMs, particularly GPT-4, can potentially assess whether an LLM-generated solution fulfills the requirements of a task and contains pertinent information. While these results are encouraging, it's crucial to consider the limitations highlighted in the paper when designing LLM-based evaluation structures. More subjective criteria, such as engagement or nuanced aspects of relevance, still pose challenges for LLMs [33].

4.11 Research Gap

While these initial findings from automated thematic analysis with LLMs, particularly for applying thematic codes and generating the codebook for inductive thematic analysis (section 4.8), are encouraging, there remains a significant research gap in our understanding of LLM capabilities within thematic analysis. Current limitations primarily revolve around two key areas:

First, existing research has primarily focused on the ability of LLMs to generate thematic structures (themes and codebooks) that resemble those produced by human researchers. However, this does not necessarily equate to the depth and nuance of findings derived from human analysis, which involves extensive engagement with the data and complex contextual understanding. For example, finding out

that the data includes a theme called "Understanding of the Flu's severeness" is not as valuable as understanding that, for example, specifically, two of the interviewees were not aware that the flu is different from a common cold. We need to understand if LLMs can generate these more nuanced findings as well.

Second, while it seems straightforward that using structured step-by-step approaches can improve the performance of human researchers in analyzing qualitative data, it needs to be analyzed if this has similar implications for how AI can best perform qualitative research. Therefore, the influence of different prompting techniques on the quality and depth of LLM-driven thematic analysis remains largely unexplored. It is unclear whether simple zero-shot prompting can achieve comparable results to approaches that guide the LLM through a structured process, such as the six steps of thematic analysis outlined by [7].

This research aims to address these gaps:

- **Comparing Prompting Techniques:** We will rigorously evaluate the impact of different prompting techniques, ranging from zero-shot approaches to structured prompting techniques that closely follow the steps of thematic analysis and incorporate prompt engineering principles.
- **Analyzing both Thematic Structures and Findings:** Going beyond comparing thematic frameworks, we will analyze the actual *results* and insights presented in the evaluation sections of LLM-generated analyses, comparing them to human-generated outputs.

Additionally, we want to evaluate if LLMs can be used to evaluate their own thematic analysis results, potentially automating not just the generation, but also the evaluation process.

- **Evaluating LLMs as Evaluators:** We will investigate whether LLMs can accurately assess the quality of their own thematic analysis results, comparing their evaluations to those of human researchers

This comprehensive approach will provide valuable insights into the potential and limitations of LLM-driven thematic analysis, paving the way for more effective integration of these powerful tools into qualitative research.

5 METHODOLOGY

5.1 Research Design

To address the identified research gap, we selected three publicly available datasets for which corresponding thematic analyses had been published in peer-reviewed research papers. This allowed us to compare the performance of our LLM-driven thematic analyses to those conducted by human researchers. Our research design consists of two distinct steps: a *generation* step and an *evaluation* step (figure 1,2).

In the generation step (figure 1), we employed different prompting techniques to generate thematic evaluations from the selected datasets. For each technique, the LLM was prompted with the original research question, the complete interview data, and instructions to write an evaluation section for a thematic analysis, including approximately three themes, each with three to five subthemes. Each model is prompted to generate the same amount of themes and subthemes to achieve comparable results. The output of this generation step was a text file containing the results section of a thematic analysis.

The subsequent evaluation step (figure 2) focused on assessing how closely the findings generated by the LLM aligned with those identified by the human researchers in the original publications. To facilitate this comparison, we developed a rubric scoring table for each dataset. These tables listed the concise key findings from the original papers, allowing us to systematically assess whether each finding was captured

by the LLM-generated analyses. Furthermore, we explored the possibility of using another LLM to fill out the rubric scoring table, effectively leveraging an LLM to evaluate the output from the generation LLM. This AI-powered evaluation approach offered a potential pathway for automating the assessment process, enabling the analysis of a larger set of LLM-generated outputs.

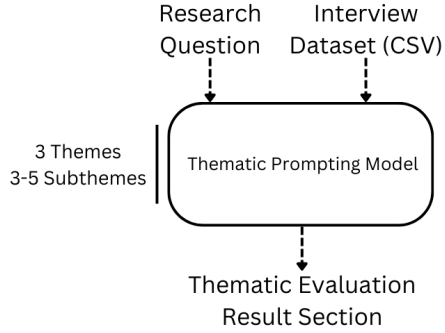


Figure 1: Generation Model Overview

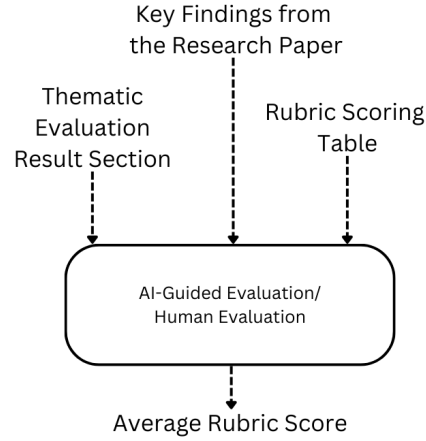


Figure 2: Evaluation Model Overview

5.2 Datasets

As a baseline for our research we have identified 3 distinct peer reviewed papers. Each of the 3 papers performed a thematic analysis on publicly accessible Interview Data, enabling us to run the SPT with the Data and then compare the findings of the SPT to those of the researchers.

5.2.1 Barriers and Facilitators of Childhood Flu Vaccination (FLU Dataset). This dataset comes from a study published in the Journal of Public Health [37], which explored "parental perspectives on childhood flu vaccination in North East England". The researchers conducted semi-structured interviews with twelve parents, six of whom had vaccinated children and six of whom had unvaccinated children. They investigated the research question: *What are the barriers and facilitators influencing the uptake of seasonal influenza vaccinations among preschool-aged children in North East England, from the perspective of parents?* Utilizing an inductive thematic analysis approach, the authors followed a systematic process of coding, theme development, and evaluation. Their analysis was informed by the COM-B model of health behavior change, resulting in a thematic framework that categorized barriers and facilitators to vaccination within the domains of capacity, opportunity, and motivation [30]. We therefore added to the research question to analyze the data from the perspective of the COM-B framework, with a short explanation of the Model (Appendix A1). This dataset provides a rich and nuanced understanding of parental decision-making regarding flu vaccination, making it a suitable testbed for evaluating the LLM's ability to extract meaningful themes from complex interview data.

5.2.2 Factors that Influence Data Sharing Through Data Sharing Platforms: A Qualitative Study on the Views and Experiences of Cohort Holders and Platform Developers (DS Dataset).

This dataset comes from a study by [Devriendt et al. \[15\]](#), which used inductive thematic analysis to explore factors influencing data sharing. The researchers conducted semi-structured interviews with seventeen participants, investigating the research question: *What factors influence data sharing decisions among cohort holders and platform developers, and how do these factors impact the utilization and success of data sharing platforms?* They coded the data inductively into themes directly derived from the content without pre-existing categories, identifying key factors such as credit and recognition, misuse concerns, control issues, and ethical barriers. These themes were iteratively refined and categorized, revealing underlying values like trust, transparency, and reciprocity that guide data-sharing behaviors. Ironically, the researchers forgot to acquire consent from all the participants to share the data and therefore the dataset we work with is only limited to 7 participant interviews.

5.2.3 A Qualitative Exploration of Advantages and Disadvantages to Using Technology in the Process of Randomised Controlled Trial Recruitment (RCT Dataset). This dataset is from a study by [Muldowney et al. \(2024\)](#) that is currently under review for submission. The study explored the "advantages and disadvantages of using technology in the recruitment process for randomized controlled trials" (RCTs) [32]. The researchers conducted semi-structured interviews with seven researchers involved in RCT recruitment and followed up with a focus group to deepen their findings. They investigated the research question: *How does the use of technology impact the recruitment process in randomized controlled trials (RCTs), particularly in terms of its advantages, disadvantages, and the factors that influence its effectiveness?* The study employed reflexive thematic analysis, an iterative and reflective process that involved coding, identifying patterns, and refining themes. This approach was particularly valuable as reflexive thematic analysis datasets are challenging to find in the public domain. The analysis revealed themes related to the practicality, costs, and context-specific success of using technology, emphasizing that it is often employed when it is perceived as convenient and resource-efficient.

5.3 LLM Generation: Thematic Prompting Techniques

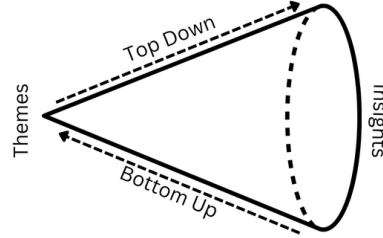


Figure 3: Visualization of Bottom Up versus Top Down Thematic Prompting Techniques

This section outlines the three prompting techniques employed for thematic analysis, each offering a distinct approach to guide the LLM in extracting meaningful insights from interview data. We begin with the most straightforward method, zero-shot prompting, which relies solely on the LLM’s pre-trained knowledge and the provided data. Next, we introduce the Top-Down Structured Prompting Technique (TDSP), which strategically leverages the LLM’s strengths in pattern recognition and summarization [39] by starting with broad themes and progressively refining the analysis. This approach draws inspiration from Chain-of-Thought (CoT) prompting and decomposed prompting, which has demonstrated improved reasoning capabilities in LLMs [23, 43]. Finally, we present the Bottom-Up Structured Prompting

Technique (BUSP), designed to mirror the meticulous and iterative process of human-driven thematic analysis as outlined by Braun and Clarke [7]. This approach, grounded in established qualitative research methods, ensures a thorough and systematic analysis firmly rooted in the data. BUSP also incorporates core LLM agent concepts like tool use, context optimization, and decision-making, to enhance each stage of the process, taking cues from recent advancements in LLM agent design [12, 29]. The difference between TDSP and BUSP is essentially that TDSP moves from themes to key insights and vice versa BUSP moves from key insights to themes as illustrated in Figure 3.

TDSP and BUSP are visualized in Figures 4 and 5, respectively, where the corresponding step of thematic analysis (section 4.3) by Braun and Clarke is noted in square brackets.

5.3.1 Zero-Shot Prompt. The zero-shot prompting technique is the most straightforward approach among the methods employed for inductive thematic analysis. In this technique, the Large Language Model (LLM) is provided with the preprocessed data in the form of CSV files and the corresponding research question. The LLM is then prompted to generate a thematic analysis without any prior structuring or intermediate steps, relying solely on its pre-trained knowledge and the input data as defined in Figure 1. Zero-shot prompting, while simple to implement, often struggles with tasks requiring complex reasoning or domain-specific knowledge [40].

While few-shot prompting has demonstrated its ability to enhance zero-shot prompting in certain contexts [11], initial trials showed no significant improvement when applying examples to our thematic analysis prompt. We hypothesize that this is because few-shot prompting primarily benefits tasks by providing the model with clearer instructions and demonstrations of the desired output. Thematic analysis, however, is a well-established principle in qualitative research, and its core concepts are likely already present within the LLM’s training data. Therefore, the additional instructions provided by few-shot examples may be redundant or less impactful in this specific case.

Therefore we only used Zero Shot prompting. The prompt used for this technique instructs the LLM to directly produce the Evaluation Section of the thematic analysis. It specifies that the output should be structured into a narrative with clear themes and subthemes, ensuring a coherent and logical flow. The prompt explicitly guides the LLM to introduce each theme, analyze its subthemes in depth, and conclude with a summary that ties the findings back to the research question. The structure is designed to include approximately three main themes, each with around five subthemes, aligning with Braun and Clarke’s recommendations for a reasonable number of themes and subthemes [7].

5.3.2 Top-Down Thematic Analyzer. The Top-Down Structured Prompting Technique (TDSP) offers a structured approach for guiding Large Language Models (LLMs) through the complex process of thematic analysis. This technique, termed “Top-Down” due to its initiation with overarching themes and subsequent refinement to granular insights (figure 3), effectively reverses the conventional thematic analysis steps outlined by Braun and Clarke [7]. TDSP draws inspiration from the success of Chain-of-Thought (CoT) prompting and decomposed prompting techniques, which have demonstrated significant improvements in LLM reasoning capabilities [23, 43]. The underlying principle lies in decomposing intricate tasks into smaller, more manageable sub-tasks, thereby unlocking greater LLM potential. By mirroring this approach and reversing the traditional thematic analysis flow, TDSP provides a structure that aligns particularly well with the strengths of LLMs.

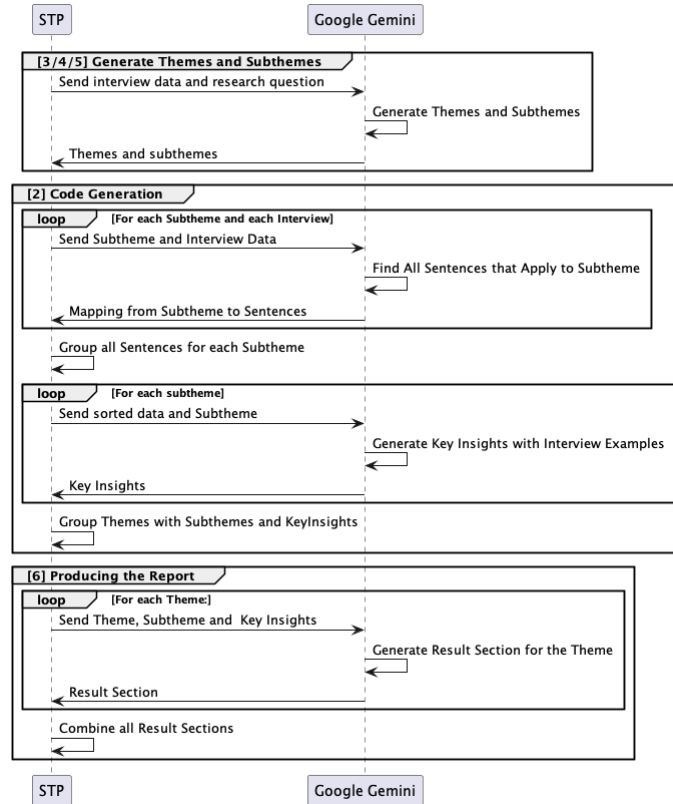


Figure 4: Top Down Thematic Analyzer

The TDSP process, visually represented in Figure 4, commences by prompting the AI with the research question and the complete interview data to generate overarching themes and subthemes (corresponding to Braun & Clarke Steps 3, 4 & 5, list 4.3). This initial prompt also includes definitions of themes and subthemes, carefully tailored to reflect the type of inductive thematic analysis employed in the original research (inductive or reflexive). This ensures alignment with the study's methodological and theoretical framework. For instance, an solely inductive approach would necessitate a definition focused on identifying patterns grounded in the data, whereas a reflexive approach would emphasize themes as representations of core underlying ideas rather than mere summaries. The LLM is then tasked with generating a concise name and a detailed description for each theme and subtheme.

These generated subthemes are then used to systematically categorize the interview data. This involves parsing the data, labeling each interviewee sentence with a unique ID (excluding interviewer questions), and prompting the AI to map these sentences to their corresponding subthemes. Finally, the data is further processed to extract all paragraphs associated with each subtheme, along with their corresponding interviewer questions. Building upon this categorized data, the AI is prompted with all the paragraph-question pairs for one subtheme and the subthemes definition to extract key insights, comparable to thematic codes in Braun and Clarke's framework (Step 2). The AI is encouraged to identify insights corroborated by multiple interviewees and to provide supporting quotes for each insight. These quotes

are crucial for bolstering the final report, with a higher frequency of quotes signifying greater prevalence and contributing to stronger argumentation. In the final stage, all the extracted information themes, subthemes, key insights, and supporting quotes are synthesized. The AI is then prompted to write a comprehensive evaluation, effectively mirroring Step 6 of Braun and Clarke’s framework, generating the evaluation for each theme independently.

The structure of TDSP offers several advantages for leveraging LLMs in thematic analysis. The top-down approach provides clear, step-by-step guidance, preventing the LLM from being overwhelmed by the intricacies of the data. This structured process, coupled with the focused provision of information at each stage (e.g., subtheme definitions and corresponding interview responses), ensures context optimization, which is crucial for maximizing the LLM’s analytical performance [23, 26]. Furthermore, TDSP aligns seamlessly with the inherent capabilities of LLMs. The initial theme and subtheme generation phase capitalizes on the LLM’s proficiency in pattern recognition and concise summarization of large datasets [39]. By minimizing noise and presenting information in a structured and focused manner, TDSP allows the LLM to fully leverage its potential for insightful thematic analysis.

5.3.3 Bottom-Up Thematic Analyzer. The Bottom-Up Structured Prompting Technique (BUSP) takes a different approach compared to the TDSP. Instead of primarily leveraging the AI’s inherent strengths in pattern recognition and not following the steps of TA in order, BUSP focuses on closely replicating the step-by-step process of human-driven thematic analysis, as outlined by Braun and Clarke (section 4.3) [7]. This “Bottom-Up” approach, starting with the generation of detailed thematic codes (similar to key insights) from the interview data and building up to the final thematic evaluation (figure 3), ensures that each stage of the analysis is firmly grounded in the data and the insights derived from it. The BUSP technique is further enriched by integrating core LLM agent concepts like tool use, context optimization, and decision-making—to enhance each stage of the process, taking cues from recent advancements in LLM agent design [12, 29]. With this fusion of human-inspired methodology and LLM agent capabilities we hope to achieve better accuracy in AI driven TA..

The BUSP is visualized in figure 5. In the first stage, mirroring Step 2 of Braun and Clarke’s framework, the LLM generates thematic codes by identifying key aspects of the interview data relevant to the research question. This process of systematically coding the data, much like a human researcher would do, ensures that the analysis is anchored in the participants’ own words and perspectives [8]. The LLM is then prompted to refine these codes by combining similar or overlapping ones, creating distinct and meaningful codes that serve as the foundation for further analysis. This refinement step exemplifies the decision-making abilities of LLM agenes, allowing the model to actively participate in shaping the analysis.

Next, the LLM applies these refined codes to the interview data, labeling each sentence according to the relevant codes, based on the same approach used to apply subthemes to the interview data in the TDSP. This step is crucial for organizing the data, as it associates specific question and answer paragraphs with appropriate thematic codes, facilitating the extraction of insights in subsequent stages.

In the third stage, the LLM uses the labeled sentences to generate key insights for each code. Key insights like in the TDSP included examples from the interviews as well as a description of the key insight. Particularly the examples are important for later writing a in the data grounded results section. Furthermore during this process, the LLM evaluates the importance of each key insight based on several

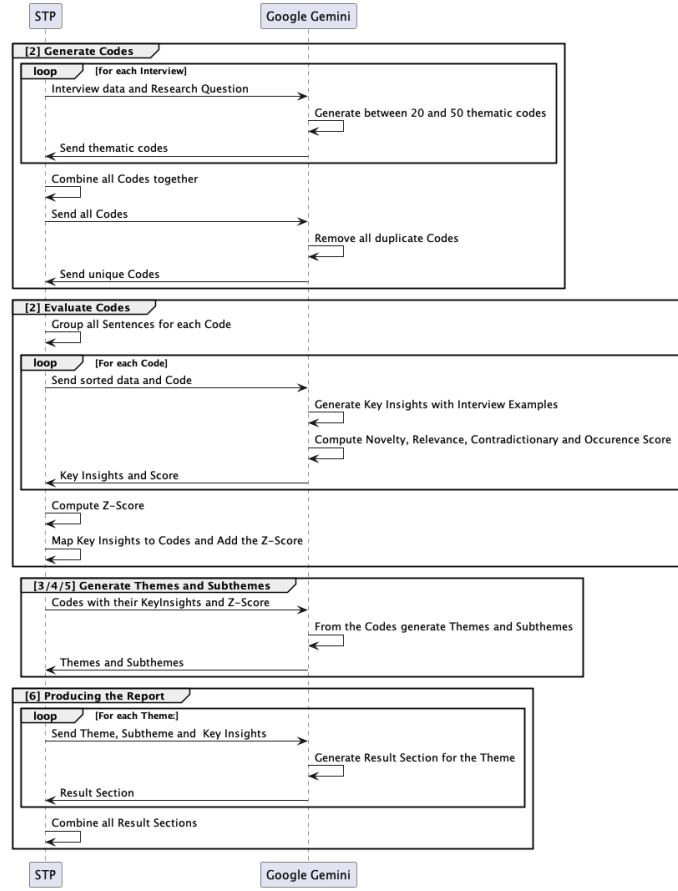


Figure 5: Bottom Up Thematic Analyzer

criteria, including the number of occurrences across different interviews, relevance to the research question, novelty, and the presence of contradictions in the data. These insights are scored on an ordinal or numerical scale, providing a comprehensive assessment that includes:

- **Occurrences:** The count of how many different interviews the insight was found in.
- **Relevance Score:** A score from 1 to 4, where 1 indicates the insight is very relevant to the research question, and 4 indicates it is not relevant.
- **Novelty Score:** A score from 1 to 4, where 1 indicates the insight is very novel, and 4 indicates it is not novel.
- **Contradictory Score:** A score from 1 to 4, where 1 indicates no contradictions in the data, and 4 indicates major contradictions.

These scores are later normalized and combined into a single Z-score, which serves as an indicator of the relative importance of each code (and its associated key insights) in contrast to others. Z-score normalization standardizes the four scoring dimensions, transforming them to have a mean of zero and a standard deviation of one, thus enabling direct comparison and aggregation into a single composite score representing the overall importance of each code [21]. This step clearly demonstrates the concept

of tool use within BUSP, leveraging the LLM’s ability to process and quantify data to create a meaningful metric for evaluating the significance of codes.

Following the generation of key insights and their respective evaluations, the LLM moves to combine the codes into broader themes and subthemes, using the appropriate definitions of inductive or reflexive themes to align with the original research paper’s methodology. The AI uses the code, its definition, its respective key insights, and the Z-scores as the basis for identifying themes and subthemes. This step corresponds to Step 3/4/5 of Braun and Clarke’s framework (section 4.3): Identifying, Refining and Defining Themes. The LLM is prompted to organize the existing codes and their key insights into approximately three overarching themes, each with between 3 and 5 subthemes. The Z-scores provide an additional metric for determining the relative importance of each code, guiding the LLM in constructing a coherent thematic structure.

Finally, the LLM synthesizes these themes, subthemes, codes, and key insights into a comprehensive thematic evaluation, as in step 6 of TA (list 4.3). The evaluation is structured to provide a detailed narrative for each theme, integrating the subthemes and grounding the analysis in the data by drawing on the key insights and examples from the interviews. This process ensures that the final output is not only insightful and well-supported by the data but also rigorously constructed, adhering to the structured approach of TA principles.

5.4 Evaluation Model: Evaluation of Findings

Now we move to the next step the evaluation of the outputs created by the generation step. Herefore, we compared the LLM-generated outputs to the original research findings presented in the publications that were released for the selected datasets (figure 2).

5.4.1 Rubric Development and Human Evaluation. To facilitate a grounded comparison, we first derived a rubric scoring table [9]. For each of the three datasets, we thoroughly reviewed the original research publication and identified approximately ten key findings, focusing on novelty and relevance to the research question. These findings served as the basis for developing our detailed rubric scoring tables. Each table outlined specific criteria for classifying an LLM-generated output as “proficient,” “partially proficient,” or “not proficient” in capturing each key finding. This approach aligns with established best practices in rubric-based assessment, ensuring a structured and consistent evaluation of complex outputs [9, 31]. The full rubrics tables are available in an online repository on github [44] and in Appendix A4/5/6. This is an example from the FLU Dataset:

- **Finding:** Active collaboration is the preferred mode of data sharing among scientists.
- **Not Proficient:** Fails to identify active collaboration as the preferred mode of data sharing. Misinterprets the preference for collaboration.
- **Partially Proficient:** Mentions active collaboration but doesn’t emphasize it as the preferred mode of data sharing. Lacks specific evidence.
- **Proficient:** Clearly states that active collaboration is the dominant preference for data sharing. Provides strong supporting evidence from the text.

From the pool of generated thematic evaluations, we selected one output per prompting technique per dataset, resulting in a total of nine papers for analysis. To mitigate potential bias, we adopted a blinding procedure. Each selected paper was labeled with a unique identifier, concealing the prompting technique used during the generation phase (Appendix A2). While this blinding method proved effective for the

TDSP and BUSP outputs, which exhibited similar lengths, the consistently shorter outputs generated by the zero-shot prompting technique remained identifiable for the Author as Evaluator.

To further mitigate potential author bias and the limitations of the blinding procedure, we enlisted an independent researcher with expertise in qualitative research to evaluate the nine selected papers. The researcher received the rubric scoring tables and the blinded LLM outputs, ensuring they were unaware of the prompting techniques used to generate each output. This independent evaluation allowed us to calculate inter-rater reliability scores, comparing the author’s assessments to those of the independent researcher, to determine the consistency and reliability of the scoring process [19]. We employed Cohen’s Kappa, a commonly utilized statistical metric for evaluating inter-rater reliability in categorical data, to assess the agreement between the two evaluators [24]. Furthermore, we computed Spearman’s rank correlation coefficients to analyze the strength and direction of the association between the scores given by the two raters [42].

5.4.2 AI Self-Evaluation. In addition to human evaluation, we explored the feasibility of using another LLM as an evaluator, which could potentially enable the analysis of a larger set of AI-generated papers. For this self-evaluation experiment, we employed the ChatGPT-4o model, recognized for its strong performance in reading comprehension and reasoning tasks, including rubric-based evaluation [2, 33, 48].

We provided ChatGPT-4 with each row of our rubric scoring table individually and prompted it to evaluate a corresponding LLM-generated output based on those criteria. By combining the results for each row, we obtained a complete rubric score for each LLM output, which we could then compare to the human-generated scores, through the inter-rater reliability score [24]. This process allowed us to directly assess the agreement between human evaluators and the AI evaluator in assessing the quality of LLM-generated thematic analyses.

However, our analysis revealed a significant discrepancy between the scores assigned by the human evaluators and those generated by the AI evaluator. This lack of agreement indicated that, despite its capabilities in other domains, the AI evaluator was not reliable for independently evaluating the quality of LLM-generated thematic analyses. While we initially intended to utilize the AI evaluator to analyze a larger set of outputs and potentially compare the performance of different prompting techniques, this lack of agreement between human and AI assessments rendered those results unreliable. Nonetheless, we include a presentation and discussion of these findings in the Results section to illustrate the challenges and limitations of AI-driven evaluation in the context of qualitative research.

6 RESULTS

This section presents the findings from our two-pronged evaluation approach, comparing the performance of three distinct prompting techniques—Zero-Shot, Top-Down Structured Prompting (TDSP), and Bottom-Up Structured Prompting (BUSP)—in generating insightful thematic analyses from qualitative datasets. Our evaluation involved both human assessments of nine LLM-generated analyses (one per prompting technique per dataset) and an exploration of using an LLM (ChatGPT-4) as an automated evaluator on a larger set of 36 outputs.

Human evaluation, conducted independently by both the author (A) and an external qualitative researcher (E) using a predefined rubric scoring system, demonstrated a “substantial” degree of inter-rater reliability [24]. We observed a Cohen’s Kappa of 0.6365 and a Spearman correlation of 0.8142 ($p < 0.001$), indicating strong agreement between the two human evaluators (Table 2). This robust agreement

provided confidence in using their assessments for further analysis. Focusing on these human judgments, we found that both structured prompting techniques, TDSP and BUSP, significantly outperformed the Zero-Shot approach. Interestingly, the Zero-Shot approach consistently produced significantly shorter outputs, averaging 1979.75 tokens, compared to the TDSP and BUSP methods, which averaged 4080.5 and 3573.83 tokens, respectively (Table 1) and we were not able to generate longer outputs through changing the prompt for the Zero Shot approach.

In contrast to the high agreement observed between human evaluators, our exploration of AI-powered evaluation revealed notable discrepancies between the assessments generated by ChatGPT-4 (AI) and those of our human evaluators. As shown in Table 2, the agreement between the AI evaluator and both human evaluators was significantly lower, with Cohen’s Kappa scores ranging from 0.1364 to 0.1489. While a detailed analysis of these differences will be presented later in this section, this initial observation, coupled with the robust agreement between human evaluators, raises concerns about the reliability of using current LLMs for the independent assessment of qualitative research outputs. We present a detailed examination of these findings, including specific examples of discrepancies and a discussion of their potential implications, in the following sections.

Algorithm	Average Tokens
ZS	1979.75
TDSP	4080.5
BUSP	3573.83

Table 1: Average Token Output for Different Algorithms

Metric	Participant Comparison	Score	P-value
Cohen’s Kappa	A vs. AI	0.1364	-
Cohen’s Kappa	E vs. AI	0.1489	-
Cohen’s Kappa	A vs. E	0.6365	-
Spearman Correlation	A vs. AI	0.3094	0.0062
Spearman Correlation	E vs. AI	0.3456	0.0021
Spearman Correlation	A vs. E	0.8142	2.1727e-19

Table 2: Cohen’s Kappa and Spearman Correlation Scores between Author (A), External Evaluator (E), and AI

6.1 Human Evaluation Results

To assess the quality of the LLM-generated thematic analyses, we selected nine outputs (one per prompting technique per dataset) for in-depth human evaluation. Both the author and an independent researcher with expertise in qualitative research evaluated these outputs using our predefined rubric scoring tables. The inter-rater reliability between the two evaluators was calculated to assess the consistency of their judgments. We observed a Cohen’s Kappa of 0.6365, indicating substantial agreement, and a Spearman correlation of 0.8142 ($p < 0.001$), further supporting the strong agreement between the two evaluators [19, 24, 42]. This high level of agreement provides confidence in the reliability and validity of our human evaluation results, allowing us to now analyze how the humans evaluated the different results from the different prompting techniques.

The assigned proficiency scores for the nine LLM-generated thematic analyses from the generation LLM revealed a clear advantage for structured prompting techniques in accurately capturing key findings from the qualitative datasets. Both Top-Down Structured Prompting (TDSP) and Bottom-Up Structured Prompting (BUSP) significantly outperformed the Zero-Shot approach, achieving mean proficiency scores of 1.55 and 1.51, respectively, compared to 1.25 for Zero-Shot (Figure 6). These scores, ranging from 0 (not proficient), 1 (partially proficient) and 2 (proficient), represent the average performance across 10 key findings identified for each dataset, with higher scores indicating greater proficiency in capturing these findings.

Examining the distribution of proficiency scores across all datasets and prompting techniques reveals a key insight: structured prompting techniques consistently resulted in a higher proportion of "Proficient" scores compared to the Zero-Shot approach (Table 3). On average, 67% of the key findings were identified as "Proficient" in both the TDSP and BUSP outputs, compared to only 48% in the Zero-Shot outputs. Furthermore, the Zero-Shot approach had a significantly higher proportion of "Not Proficient" scores (22%) compared to TDSP (12%) and BUSP (16%).

Algorithm	Not Proficient	Partially Proficient	Proficient
SH	0.22	0.29	0.48
TD	0.12	0.21	0.67
BU	0.16	0.17	0.67

Table 3: Average Number of Scores for the Combined Datasets

These findings strongly suggest that guiding LLMs through a structured, step-by-step process significantly improves their ability to identify and accurately capture key findings in thematic analysis. Among the structured techniques, TDSP emerged as the most consistent performer, achieving the highest average proficiency score across all datasets and consistently yielding a high proportion of "Proficient" scores.

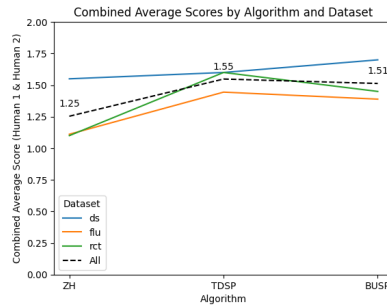


Figure 6: Mean Proficiency Score Human Evaluators

6.2 In-Depth Analysis of FLU Dataset Findings

The human evaluation of LLM-generated thematic analyses on the FLU dataset revealed both strengths and limitations in capturing key findings related to the flu vaccination dataset. As shown in Figure 7(a), TDSP emerged as the most successful prompting technique, achieving a "Proficient" score (as assigned

by both evaluators) for six out of the nine key findings. BUSP achieved "Proficient" scores for four of the findings, while Zero-Shot only reached this level of proficiency for three.

Notably, all prompting techniques failed to identify Finding 1, which focused on the association between socioeconomic factors and vaccination uptake.

Zero-Shot prompting consistently struggled to capture more complex or subtle findings. Specifically, Findings 2, 7, and 9, which addressed parental knowledge, social norms, and negative experiences with healthcare providers, respectively, were not adequately identified by the Zero-Shot approach. In contrast, at least one of the structured prompting techniques (TDSP or BUSP) achieved "Proficient" scores for these findings.

Finally, there were some disagreements between the two human evaluators regarding specific outputs. For example, on Finding 7 (social norms) and Finding 9 (negative experiences), the evaluators disagreed on the proficiency level of the BUSP output.

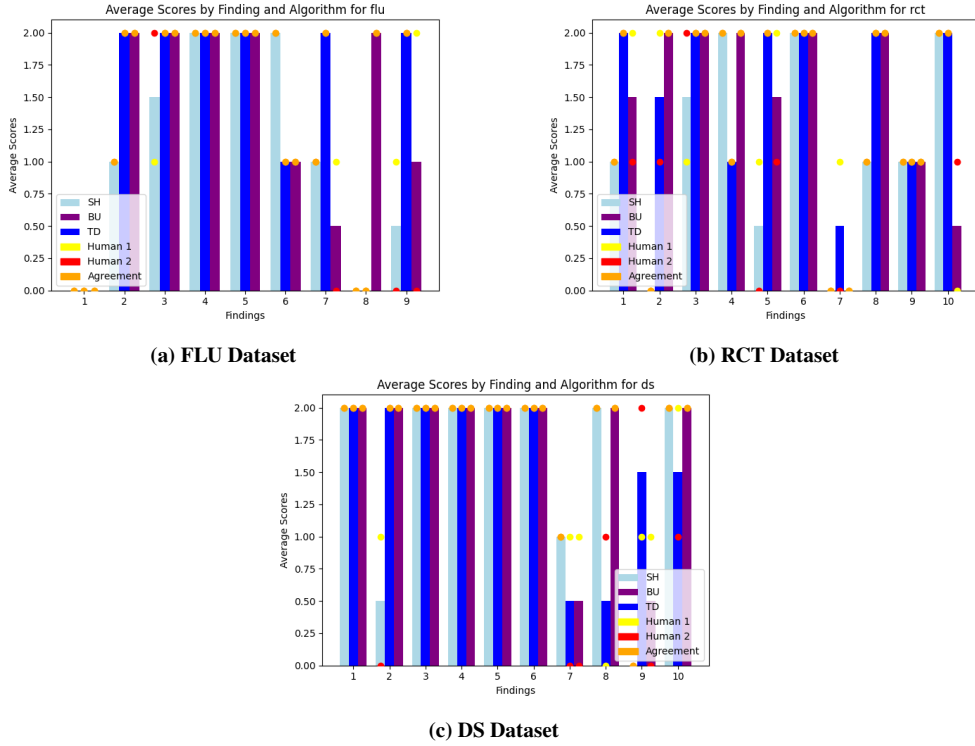


Figure 7: Results for each finding for the different prompting techniques for the different datasets

6.3 In-Depth Analysis of RCT Dataset Findings

Human evaluation of LLM-generated thematic analyses on the RCT dataset, which explored the role of technology in randomized controlled trial recruitment, showed a high overall proficiency across prompting techniques. However, variations in performance emerged when examining individual findings. As illustrated in Figure 7(b), both TDSP and BUSP achieved "Proficient" scores for six out of the ten key

findings. Zero-Shot obtained "Proficient" scores for five of the findings. Among the three prompting techniques, TDSP emerged as the most successful, achieving the highest average proficiency score (1.55) and consistently capturing key themes across the dataset.

All prompting techniques consistently identified Findings 3, 6, and 9. This suggests that these themes, related to participant burden reduction, the potential impact of technology on diversity, and the need to consider technology preferences beyond generational assumptions, were relatively straightforward for the LLMs to identify, regardless of the prompting approach.

Both structured prompting techniques (TDSP and BUSP) outperformed Zero-Shot on Finding 4, which highlighted the importance of tailoring recruitment strategies to the target population. Similarly, structured prompting techniques were more successful than Zero-Shot in capturing Finding 8, emphasizing the importance of clear communication about data security. Structured prompting techniques also demonstrated an advantage on Findings 1, 2 and 5, which explored strategic use of technology, the influence of cost-effectiveness on technology choices, and the role of gatekeepers in recruitment.

However, all three techniques, including TDSP, struggled with Finding 7, which focused on ethical approval processes as a potential barrier. BUSP also failed to achieve a "Proficient" score for Finding 10, relating to the disconnect between researchers and patient communities.

6.4 In-Depth Analysis of DS Dataset Findings

The human evaluation of LLM-generated thematic analyses on the DS dataset, which focused on factors influencing data sharing among scientists, showed a high level of proficiency across all prompting techniques. As depicted in Figure 7(c), both TDSP and BUSP achieved a "Proficient" score for six out of the ten key findings, while Zero-Shot attained "Proficient" scores for five.

All prompting techniques consistently identified Findings 1, 3, 4, 5, 6, and 10. These results suggest that these themes were prominently represented in the dataset and relatively straightforward for the LLMs to identify, regardless of the prompting approach.

However, there were variations in performance on the remaining four findings. Zero-Shot prompting struggled to fully capture the nuances of Finding 2 (inadequate recognition for data contribution without active involvement) and Finding 9 (the role of reciprocity in ensuring fair benefit sharing). Both structured prompting techniques successfully identified these findings, highlighting their ability to delve deeper into complex themes.

Interestingly, both TDSP and BUSP had difficulty consistently capturing Finding 7, which focused on the influence of socio-cultural factors on data sharing behaviors.

On Finding 8 (the importance of trust in data sharing), both BUSP and Zero-Shot achieved "Proficient" scores, while TDSP did not.

6.5 Results of the AI Self-Evaluation

To further explore the potential of LLMs in evaluating thematic analysis, we used ChatGPT-4 as an automated evaluator, applying it to a larger set of 36 LLM-generated thematic analyses (3 prompting techniques x 3 datasets x 4 variations). Each output was scored using the same rubric system employed by our human evaluators.

However, in contrast to the high agreement observed between human evaluators, our analysis revealed notable discrepancies between the assessments generated by ChatGPT-4 (AI) and those of our human evaluators. As shown in Table 2, the agreement between the AI evaluator and both human

evaluators was significantly lower, with Cohen’s Kappa scores ranging from 0.1364 to 0.1489. These scores, which fall below the threshold of 0.1 suggest a lack of substantial concordance between the AI evaluator and human judgments. A score below 0.1 can be interpreted as no agreement between evaluators [24]. The Spearman correlation scores, although slightly higher (0.3094 to 0.3456), also indicate only a moderate, positive relationship between the rankings assigned by the AI and human evaluators [42].

A closer examination of the average proficiency scores generated by the AI evaluator (Appendix A3) reveals intriguing dataset-specific variations. In the FLU dataset, the AI evaluator assigned the highest average proficiency score (1.64) to the Zero-Shot outputs, followed by TDSP (1.39) and then BUSP (1.25). This trend directly contrasts with the human evaluations, where both TDSP and BUSP outperformed the Zero-Shot approach. Similarly, in the DS dataset, the AI evaluator again favored Zero-Shot (1.52) over both structured prompting techniques (TDSP: 1.32, BUSP: 1.42). However, in the RCT dataset, the AI evaluator’s scores aligned more closely with the human evaluations, with TDSP receiving the highest score (1.62), followed by Zero-Shot (1.52) and BUSP (1.15).

While human evaluators consistently favored structured prompting techniques across datasets, the AI evaluator displayed considerable variability in its assessments, sometimes aligning with human judgments and other times diverging significantly. This inconsistency highlights a critical limitation of using current LLMs for evaluating tasks that require nuanced understanding and interpretation, underscoring the continued importance of human expertise in qualitative research. We will further explore the potential reasons for these discrepancies and discuss their implications for the future development and the discussion section.

7 DISCUSSION

This study investigated the potential of structured prompting techniques to enhance the accuracy and depth of LLM-driven thematic analysis, comparing their performance to a simpler Zero-Shot approach and exploring the feasibility of using an LLM as an evaluator. Our findings reveal both promising advancements and persistent challenges in leveraging LLMs for qualitative research.

7.1 Interpretation of Results

Our results strongly confirm that structured prompting techniques, particularly Top-Down Structured Prompting (TDSP), significantly enhance the accuracy and depth of LLM-driven thematic analysis. Both TDSP and Bottom-Up Structured Prompting (BUSP) consistently outperformed the Zero-Shot approach across multiple datasets, achieving higher proficiency scores and capturing a greater proportion of key findings. This success likely stems from the ability of structured prompting to guide LLMs through a more methodical, step-by-step process, mimicking the stages of human thematic analysis [7] and incorporating elements of prompt engineering and context optimization.

TDSP, in particular, builds upon established prompt engineering principles, first deriving overall themes and then gradually building up key insights. This process, strongly based on decomposed prompting and context optimization [23, 26], aligns well with LLMs’ natural strengths in pattern recognition and summarization [39]. In contrast, BUSP leverages the structure that humans have developed for effectively analyzing qualitative data, prioritizing the generation of detailed codes before grouping them into themes. While this bottom-up approach also demonstrated promise, it appears that the nature of LLMs might benefit more from building up codes from themes rather than the reverse. We can identify two

reasons why moving from themes to key insights might be beneficial. Firstly, the model benefits from using the entire interview dataset as input for generating overarching themes, as in TDSP, compared to BUSP, where the themes are generated only from the reduced context of derived thematic codes and key insights. Secondly, by first generating subthemes and then labeling the sentences to which those subthemes apply, the key insights generated from this reduced context are of higher quality than those produced by generating codes directly from the raw interview data, as in BUSP. Both solutions could be tested to determine if they enhance the current state of the BUSP technique, for example, by generating all codes directly from the full dataset or, alternatively, by first subdividing the dataset into overarching themes before generating the thematic codes.

One aspect of the BUSP technique that warrants further investigation is the LLM’s ability to score key insights based on the five criteria (occurrences, relevance, novelty, contradictions, and quotes). While our study design did not allow us to definitively assess whether this scoring mechanism improved the technique’s overall performance, the low reliability of the AI evaluator in our separate evaluation suggests that similar limitations might apply to the LLM’s scoring within BUSP. Further research is needed to explore the impact of this scoring mechanism on BUSP’s effectiveness and to potentially refine the criteria or scoring process.

The dataset-specific variations in prompting technique performance further highlight the complex interplay between the nature of the data, the specific research questions, and the strengths and limitations of different prompting approaches. The FLU dataset [37], which explored parental decision-making regarding childhood flu vaccination, proved particularly challenging for the Zero-Shot approach, suggesting that capturing the nuances of personal beliefs, experiences, and motivations might require more structured guidance for LLMs. The DS dataset [15], focusing on data sharing practices among scientists, revealed the potential of BUSP to outperform TDSP in certain contexts, potentially when the themes are highly interconnected. The RCT dataset [32], which examined the role of technology in recruitment, highlighted the consistent performance of TDSP across a range of findings, further emphasizing the value of a top-down approach for analyzing strategic decision-making processes. These variations suggest that the optimal prompting technique for LLM-driven thematic analysis might be contingent upon the specific characteristics of the dataset and the research questions being explored, highlighting that both techniques have unique strengths. For improving the performance of STPs in the future, it will be crucial to better understand these strengths and weaknesses.

Despite the promise of LLMs in performing thematic analysis, our exploration of using ChatGPT-4 as an automated evaluator exposed a critical limitation of current AI capabilities. The AI evaluator demonstrated low agreement with human judgments, as evidenced by low Cohen’s Kappa scores and only moderate Spearman correlations. This finding, juxtaposed with the robust agreement observed between our human evaluators, highlights the complexities involved in accurately assessing the quality and nuance of thematic analysis. LLMs, despite their advanced language processing capabilities, might still struggle to grasp the subtle contextual cues, interpret implicit meanings, and understand the broader research implications that inform human evaluation. It is possible that the AI evaluator, despite being provided with detailed rubrics [9, 31], over-relies on surface-level features or patterns in the text, failing to capture the deeper meaning and relationships between themes that are readily apparent to human evaluators. Moreover, the AI evaluator’s inconsistent performance across datasets suggests a potential sensitivity to the specific characteristics of the data or the research questions being explored. This inconsistency

underscores the need for careful consideration and potential calibration of AI-based evaluation systems when applied to different qualitative research contexts.

Our findings regarding the AI evaluator’s limitations are in line with existing research. [Murugadoss et al.](#) also observed only a moderate correlation between LLM-generated scores and human judgments on rubric-based evaluations, particularly for criteria related to relevance, which is central to our study [33]. Furthermore, [22] highlighted an intriguing phenomenon: rubric-based evaluation by LLMs showed good inter-rater reliability with human coders only after the LLM had learned the individual scoring tendencies of each human coder. This suggests that training an AI evaluator on the specific scoring patterns of the human evaluators involved in a study might improve its reliability. Another idea is that the LLM has to evaluate all papers at once to generate relative judgment. A preliminary exploration of having the LLM evaluate all papers simultaneously to enable relative scoring did not yield significant improvements, but further research in this direction might prove fruitful. The development of a reliable AI evaluator would be a significant benefit for future studies, enabling the efficient analysis of larger datasets and potentially accelerating the pace of research.

7.2 Implications for Qualitative Research

Our findings demonstrate that structured prompting techniques, specifically TDSP, significantly enhance the depth and nuance of insights generated by LLMs compared to Zero-Shot prompting, effectively addressing our research question. TDSP, by leveraging the strengths of prompt engineering, emerged as the most effective technique in our study. However, the promising performance of BUSP suggests that incorporating human-inspired methods holds significant potential.

Beyond simply improving the accuracy of LLM-driven analysis, structured prompting offers a critical advantage: transparency. The inherent step-by-step nature of these techniques documents the AI’s decision-making process, providing researchers with a clear understanding of how the LLM arrived at its findings and enabling them to intervene or adjust the analysis when necessary. This transparency is particularly crucial in qualitative research, where the complexities and nuances of human experiences often require careful interpretation and human oversight.

Moreover, our findings suggest that LLM-based thematic analysis, even if not yet perfectly accurate, can be a valuable tool for researchers to critically examine and potentially refine their own interpretations. The observation that certain findings were consistently missed by all prompting techniques across datasets might point to potential blind spots or areas where the original human analysis could be strengthened. In this way, LLMs can act as a “second reader,” prompting researchers to re-evaluate their assumptions, consider alternative perspectives, and ultimately contribute to more robust and valid qualitative findings.

These promising results and the remaining challenges identified in our study highlight the need for further research to fully realize the potential of AI-assisted thematic analysis. We outline specific recommendations for future research directions in the following section.

8 RECOMMENDATIONS FOR FUTURE RESEARCH

Building upon our findings, we recommend two key areas for future research:

- (1) **Fine-Grained Analysis of Prompting Techniques:** To further understand the nuanced performance variations observed across different prompting techniques, future research should focus

- on a detailed analysis of each step within the prompting process. This involves isolating and evaluating the performance of individual steps, such as theme generation, thematic code generation, and the writing of the result section, for each technique. By examining the strengths and weaknesses of each step in isolation, researchers can gain a deeper understanding of how different prompting approaches interact with the capabilities of LLMs and identify areas for optimization.
- (2) **Enhancing AI-Powered Evaluation:** The development of a reliable and robust AI-powered evaluation system remains a critical challenge for advancing LLM-driven thematic analysis. We believe that this challenge can be addressed through two key strategies:
- **Incorporating Human Scoring Preferences:** By incorporating information about the scoring tendencies of human evaluators into the AI’s prompts, we can potentially improve the alignment between AI-generated scores and human judgments. This could involve training the AI on a corpus of human-annotated outputs, allowing it to learn the subtle preferences and decision-making patterns of individual evaluators.
 - **Enabling Relative Scoring:** Instead of evaluating each LLM output in isolation, future research should explore providing the AI with the ability to score outputs relative to one another. This could involve presenting the AI with multiple outputs for the same finding simultaneously, allowing it to make comparative judgments and assign scores that are more consistent and aligned across different outputs.

By pursuing these research avenues, we can further refine and enhance LLM-driven thematic analysis, moving closer to a future where AI tools effectively support and collaborate with human researchers in unraveling the complexities of qualitative data.

9 CONCLUSION

This study has demonstrated the significant potential of structured prompting techniques, particularly TDSP, to enhance the accuracy and depth of LLM-driven thematic analysis. The success of TDSP, which aligns with established prompt engineering principles and leverages the inherent strengths of LLMs, offers a promising pathway for incorporating AI into qualitative research practices. While the promising results of BUSP suggest that integrating human-inspired methods also holds value, further research is needed to refine this approach and explore its applicability to different types of qualitative data.

Our exploration of AI-powered evaluation revealed significant limitations in the ability of LLMs to independently assess the quality of thematic analyses. However, the insights gained from this exploration, coupled with the successful implementation of structured prompting techniques, provide a valuable foundation for future research aimed at developing more robust and reliable AI-powered tools for qualitative research.

While our findings are encouraging, there is still significant room for improvement in LLM-driven thematic analysis. We believe that structured prompting techniques, particularly those that prioritize transparency and provide researchers with a clear understanding of the LLM’s reasoning process, represent the most promising path forward. This transparency not only enables researchers to intervene and guide the analysis when necessary but also fosters greater trust and confidence in the results, paving the way for a more collaborative and insightful future for qualitative research.

Furthermore, this study has established a robust framework for evaluating the effectiveness of AI-driven thematic analysis. Our framework employs a structured prompting technique to generate result sections of qualitative research papers, which are then assessed using a predefined rubric scoring table

completed by human evaluators. By analyzing inter-rater reliability between human evaluators, we ensured the validity and consistency of the scoring process. This framework allows for direct comparisons between different prompting techniques and the original human-generated analyses, providing a robust and quantifiable measure of the LLM’s ability to capture key findings. Future research can leverage this framework to analyze a wider range of datasets, incorporating more research papers and involving additional human evaluators to further strengthen the analysis. This standardized approach will facilitate a more comprehensive understanding of the capabilities and limitations of AI-driven thematic analysis and its potential to transform qualitative research practices.

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APPENDIX

A1: Research Questions

- **Barriers and Facilitators of Childhood Flu Vaccination:** What are the barriers and facilitators influencing the uptake of seasonal influenza vaccinations among preschool-aged children in North East England, from the perspective of parents? This question will be examined through the lens of the COM-B model of health behavior change, focusing on Capability, Opportunity, and Motivation.
- **Factors Influencing Data Sharing Decisions:** What factors influence data sharing decisions among cohort holders and platform developers, and how do these factors impact the utilization and success of data sharing platforms?

- **Impact of Technology on RCT Recruitment:** How does the use of technology impact the recruitment process in randomized controlled trials (RCTs), particularly in terms of its advantages, disadvantages, and the factors that influence its effectiveness?

Dataset	Prompting Technique	UUID	Comparison	Metric	Score
flu	ZH	CAC8C6B5	E vs. AI	Cohen's Kappa	0.28
flu	ZH	CAC8C6B5	A vs. AI	Cohen's Kappa	0.3208
flu	ZH	CAC8C6B5	E vs. A	Cohen's Kappa	0.6727
flu	TDSP	E1EECA3F	E vs. AI	Cohen's Kappa	0.0270
flu	TDSP	E1EECA3F	A vs. AI	Cohen's Kappa	0.0270
flu	TDSP	E1EECA3F	E vs. A	Cohen's Kappa	1.0
flu	BUSP	BE5F6A04	E vs. AI	Cohen's Kappa	0.1628
flu	BUSP	BE5F6A04	A vs. AI	Cohen's Kappa	0.3077
flu	BUSP	BE5F6A04	E vs. A	Cohen's Kappa	0.6087
rect	ZH	8A48E84F	E vs. AI	Cohen's Kappa	0.1304
rect	ZH	8A48E84F	A vs. AI	Cohen's Kappa	0.0769
rect	ZH	8A48E84F	E vs. A	Cohen's Kappa	0.7015
rect	TDSP	96E2BE60	E vs. AI	Cohen's Kappa	-0.4286
rect	TDSP	96E2BE60	A vs. AI	Cohen's Kappa	-0.0204
rect	TDSP	96E2BE60	E vs. A	Cohen's Kappa	0.5918
rect	BUSP	8BC3F719	E vs. AI	Cohen's Kappa	0.4828
rect	BUSP	8BC3F719	A vs. AI	Cohen's Kappa	0.1803
rect	BUSP	8BC3F719	E vs. A	Cohen's Kappa	0.4915
ds	ZH	007B73A7	E vs. AI	Cohen's Kappa	-0.2000
ds	ZH	007B73A7	A vs. AI	Cohen's Kappa	-0.1111
ds	ZH	007B73A7	E vs. A	Cohen's Kappa	0.7872
ds	TDSP	58F45A95	E vs. AI	Cohen's Kappa	0.3443
ds	TDSP	58F45A95	A vs. AI	Cohen's Kappa	0.3443
ds	TDSP	58F45A95	E vs. A	Cohen's Kappa	0.1304
ds	BUSP	9EB18D6B	E vs. AI	Cohen's Kappa	0.2857
ds	BUSP	9EB18D6B	A vs. AI	Cohen's Kappa	0.1176
ds	BUSP	9EB18D6B	E vs. A	Cohen's Kappa	0.4444

Table A2: Detailed Cohen's Kappa Scores for Different Comparisons, E = External Evaluator, A = Author, AI = AI

Dataset	Algorithm	Average Proficient Score	Best Proficient Score
FLU	ZH	1.64	1.78
FLU	TDSP	1.39	1.78
FLU	BUSP	1.25	1.56
RCT	ZH	1.52	1.70
RCT	TDSP	1.62	1.70
RCT	BUSP	1.15	1.50
DS	ZH	1.52	1.60
DS	TDSP	1.32	1.40
DS	BUSP	1.42	1.50

Table A3: Average and Best Proficient Scores by Dataset and Algorithm when Evaluated with AI

ID	Finding	Not Proficient	Partially Proficient	Proficient
1	Active collaboration is the preferred mode of data sharing among scientists.	- Fails to identify active collaboration as the preferred mode. - Misinterprets the preference for collaboration.	- Mentions active collaboration but doesn't emphasize it as the preferred mode. - Lacks specific evidence.	- Clearly states active collaboration as the dominant preference. - Provides strong supporting evidence.
2	Scientists perceive inadequate recognition for data contribution without active involvement.	- Doesn't mention lack of recognition for data contribution without active involvement.	- Acknowledges concern of inadequate recognition but doesn't connect it to perceived value without involvement.	- Explicitly states concern of insufficient recognition and provides strong supporting evidence.
3	Misinterpretation of data is a significant concern related to data sharing without active involvement.	- Fails to identify misinterpretation of data as a concern.	- Mentions misinterpretation but doesn't elaborate on its impact.	- Clearly articulates misinterpretation of data as a key concern, with concrete examples.
4	Lack of resources, including time and funding, is a major barrier to data sharing.	- Doesn't recognize lack of resources as a barrier.	- Mentions lack of resources but focuses only on one constraint (e.g., time, funding).	- Clearly identifies lack of resources as a significant barrier, discussing time, funding, and administrative support.
5	Scientists express concerns about the loss of control over data once it is shared.	- Doesn't identify loss of control over data as a concern.	- Mentions the concern of loss of control but elaborates only one specific concern (e.g., unauthorized use).	- Clearly articulates the concern of losing control, providing examples of these anxieties.
6	Ethical and legal barriers, particularly related to GDPR compliance, pose challenges to data sharing.	- Fails to mention ethical and legal barriers as challenges.	- Mentions ethical and legal barriers, but addresses only one specific challenge (e.g., GDPR).	- Clearly identifies ethical and legal barriers, addressing GDPR, consent, and bureaucratic challenges.
7	Socio-cultural factors, such as scientists' desire for active research involvement, influence data sharing behaviors.	- Fails to recognize socio-cultural factors as influencing behaviors.	- Mentions socio-cultural factors but provides only general statements without specific examples.	- Clearly articulates how socio-cultural factors influence willingness to share data.
8	The establishment of trust is a prerequisite for data sharing, especially with unfamiliar collaborators or institutions.	- Fails to identify trust as a factor influencing data sharing.	- Mentions the need for trust but doesn't explain its connection to willingness to share data.	- Clearly articulates the role of trust and reluctance to share with unfamiliar entities.
9	'Reciprocity' in data sharing ensures all parties benefit fairly from data contributions.	- Fails to relate 'reciprocity' to data sharing practices.	- Mentions 'reciprocity' but doesn't elaborate on its connection to fair benefit sharing.	- Clearly explains how 'reciprocity' ensures fair benefit sharing and prevents disadvantages.
10	'Transparency' in data sharing involves clear communication regarding data access, usage, and authorship.	- Fails to connect 'transparency' to data sharing practices.	- Mentions 'transparency' but fails to elaborate on its importance for fostering trust.	- Clearly articulates how 'transparency' is essential for building trust and promoting successful collaborations.

Table A4: Rubric Scoring Table for the Data Sharing Dataset

ID	Finding	Not Proficient	Partially Proficient	Proficient
1	Socioeconomic factors, including age, deprivation level, and employment status, are associated with preschool influenza vaccination uptake.	- Fails to identify any association between socioeconomic factors and vaccination status. - Misinterprets the demographic data.	- Mentions socioeconomic factors but does not explain the specific relationships observed or their relevance to vaccination uptake.	- Explains that socioeconomic factors like age, deprivation, and employment status influence vaccination decisions.
2	Knowledge and awareness of influenza, including its severity and potential impact on children, play a key role in parental decision-making.	- Fails to mention the importance of parental knowledge about influenza.	- Mentions knowledge but does not differentiate varying levels or explain its influence on vaccination choices.	- Explains that strong knowledge of influenza influences decisions, while lack of knowledge correlates with lower vaccination uptake.
3	Parental perceptions of the safety and effectiveness of the influenza vaccine are crucial in shaping their willingness to vaccinate their children.	- Fails to mention parental perceptions of vaccine safety or effectiveness.	- Mentions safety or effectiveness but does not explain their influence on decisions.	- Explains that positive perceptions of safety and effectiveness lead to higher vaccine uptake.
4	Trust in information sources, particularly official health organizations and established medical institutions, influences parental views on influenza and vaccination.	- Fails to mention the role of trusted information sources.	- Mentions trusted sources but does not elaborate on their role in shaping decisions.	- Explains that reliance on trusted sources like the NHS strongly influences vaccination decisions.
5	Convenience and accessibility of vaccination opportunities, including appointment scheduling and vaccine availability, play a significant role in facilitating uptake.	- Fails to mention the importance of convenience and access.	- Mentions convenience or access but does not provide specific examples.	- Explains how flexible appointment times and vaccine availability significantly increase uptake.
6	Fear of their child experiencing a severe case of influenza is a strong motivator for parents to vaccinate, particularly for those with children considered high-risk.	- Fails to mention fear of severe influenza as a motivating factor.	- Mentions fear but does not link it to risk perception or concerns.	- Explains that fear of severe influenza, especially for high-risk children, is a major motivator for vaccination.
7	Social norms and a sense of collective responsibility contribute to vaccination uptake.	- Fails to address the influence of social norms or collective responsibility.	- Mentions social factors but does not explain their role in shaping choices.	- Explains that social influences and collective responsibility drive vaccination uptake.
8	Concerns about potential side effects, often based on anecdotal evidence, can act as a barrier to influenza vaccination.	- Fails to mention concerns about vaccine side effects.	- Mentions side effects but does not elaborate on their origins or impact.	- Explains that side effect concerns, often based on anecdotal evidence, act as barriers to vaccination.
9	Negative experiences with or distrust of medical professionals can impact parental motivation to vaccinate, even if it doesn't prevent vaccination altogether.	- Fails to mention the role of healthcare provider interactions.	- Mentions distrust but does not link it to vaccination motivation.	- Explains that negative past experiences with healthcare providers can reduce willingness to vaccinate, even if they don't prevent vaccination.

Table A5: Rubric Scoring Table for the Flu Dataset

ID	Finding	Not Proficient	Partially Proficient	Proficient
1	Researchers view technology as a valuable tool in recruitment but prioritize its strategic use to address specific needs.	- Fails to recognize the pragmatic approach to technology use. - Misinterprets the researchers' views on technology.	- Mentions the use of technology but does not explain the strategic rationale.	- Explains that researchers prioritize using technology to address specific challenges, not just for innovation.
2	Cost-effectiveness is a major factor influencing technology choices in recruitment.	- Fails to mention cost as a factor in technology adoption. - Misinterprets financial constraints in research.	- Mentions cost but does not elaborate on its impact or trade-offs.	- Explains that researchers prioritize low-cost technologies to minimize expenses while ensuring recruitment strategies are feasible.
3	Technology can reduce participant burden by minimizing time commitments, travel costs, and face-to-face interactions.	- Fails to identify the benefits of technology for participants. - Misinterprets the impact on participant burden.	- Mentions reducing participant burden but lacks specific examples.	- Explains how technology reduces participant burden by offering remote options for screening and data collection.
4	Recruitment strategies must be tailored to the target population, considering factors like age, technology literacy, and access.	- Fails to acknowledge the importance of aligning recruitment methods with the target population.	- Mentions the target population but does not explain how characteristics should influence strategy.	- Explains that researchers must consider demographics and preferences to develop effective and inclusive strategies.
5	Gatekeepers are increasingly vital in an era of increased data privacy regulations like GDPR.	- Fails to define gatekeepers or their role in recruitment. - Misunderstands the concept of gatekeeping in research.	- Provides a basic definition of gatekeepers but lacks emphasis on GDPR context.	- Explains that gatekeepers help connect researchers to participants ethically, especially in the context of GDPR.
6	Technology can expand recruitment reach but may inadvertently narrow sample diversity.	- Fails to acknowledge the potential negative impact of technology on sample diversity.	- Mentions sample diversity but does not explain how technology choices affect it.	- Explains that technology can expand reach but may limit diversity if not carefully planned.
7	Ethical approval processes can be perceived as a barrier by researchers.	- Fails to mention ethical approval processes or challenges.	- Mentions ethical approval but does not explain why it can be frustrating.	- Recognizes that obtaining ethical approval is essential but also highlights the challenges in navigating these processes.
8	Clear and transparent communication about data security is crucial for building trust and encouraging participation.	- Fails to mention transparency and communication regarding data security.	- Mentions data security or transparency but does not explain their importance for trust-building.	- Explains that clear communication about data security is essential for fostering trust in technology-based recruitment.
9	The assumption that younger generations are universally comfortable with technology can be problematic.	- Fails to challenge the assumption of universal comfort with technology among younger generations.	- Mentions variations in comfort levels but does not explain the implications for recruitment.	- Explains that even younger participants may need support or prefer alternative communication methods, necessitating inclusive recruitment strategies.
10	A disconnect between researchers and patient communities can hinder recruitment efforts.	- Fails to acknowledge the potential disconnect between researchers and participants.	- Mentions a disconnect but does not explain its implications for recruitment.	- Explains that engaging directly with communities helps researchers better understand their needs and build trust, improving recruitment outcomes.

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Table A6: Rubric Scoring Table for the RCT Dataset